

Predictive Linear Trajectory Control Barrier Function for 3D UAVs Under Dwell-Time Constraints

Abstract—This paper proposes a predictive linear trajectory control barrier function (PLTCBF) approach for handling dwell-time constraints in three-dimensional UAV flight. The dwell-time constraint considered here limits each continuous stay of a UAV inside a time-limited zone (TLZ), such as a radar-monitored or signal-degraded region where short traversal is allowed but prolonged stay is unsafe. Unlike strict no-fly zones (NFZs), the dwell-time constraint of the TLZ implicitly couples spatial entry with temporal duration, making the CBF construction nontrivial. To address this challenge, a predictive linear trajectory (PLT) is introduced to preview the UAV’s future motion using its current state. Accounting for the motion of the TLZ, the resulting relative PLT describes the predicted motion of the UAV with respect to the TLZ. For non-spherical TLZs, ellipsoidal TLZs are transformed into equivalent spheres, while the horizontal cross-sections of elliptic-cylinder TLZs are transformed into equivalent circles and analyzed in the horizontal plane. With these transformations, the dwell-time constraint is characterized through the geometric relationship between the PLT and the TLZ in the transformed space. Based on this geometric characterization, the PLTCBF is constructed from three component CBFs corresponding to different UAV–TLZ relations. We then analytically prove the validity of the construction, the preservation of safety under mode switching, and the resulting dwell-time guarantee. The corresponding CBF condition is incorporated into a quadratic-programming safety filter for online control. Numerical simulations confirm that the proposed PLTCBF satisfies the dwell-time constraint while reducing unnecessary detours compared with treating TLZs as strict NFZs. A closed-loop CoppeliaSim simulation further demonstrates the implementation of PLTCBF within a quadrotor control architecture.

Index Terms—Dwell-time constraints, spatio-temporal constraints, control barrier functions, unmanned aerial vehicles.

I. INTRODUCTION

During practical missions, unmanned aerial vehicles (UAVs) must satisfy various constraints while maintaining safe flight [1], [2], [3], [4], [5]. Many of these constraints involve spatio-temporal coupling, where spatial conditions and temporal limitations must be considered jointly [6], [7], [8]. This paper focuses on one representative form of such constraints, namely the dwell-time constraint induced by time-limited zones (TLZs) [9].

Specifically, the dwell-time constraint limits each continuous stay of the UAV inside a TLZ by a prescribed time limit. In signal-limited regions, the UAV should depart before communication or navigation performance falls below an acceptable threshold [10]. In hostile airspace, it should exit radar coverage before being tracked or locked on [11]. In harsh environments, the maximum dwell time may further be constrained by thermal or radiation exposure limits [12]. These examples illustrate the implicit coupling between spatial occupancy and temporal duration, which makes the dwell-time constraint difficult to handle directly. A conservative

approach is to approximate the TLZ as a strict no-fly zone (NFZ). However, this simplification ignores the temporal nature of the original constraint and may exclude otherwise feasible trajectories, thereby reducing the feasible mission space and mission efficiency.

Although the dwell-time constraint considered in this paper has received limited attention, several related forms of spatio-temporal constraints have been investigated. One related class concerns explicitly time-varying spatial constraints, such as moving-obstacle avoidance, where the temporal dependence is directly embedded in the obstacle motion [13], [14]. Model predictive control (MPC) provides an effective tool for such problems by predicting obstacle evolution and enforcing collision-avoidance constraints over a finite horizon [15], [16], [17], [18]. Another related class concerns task-level temporal requirements, such as reaching prescribed regions within specified time windows. These requirements are often formulated as constrained trajectory optimization problems, where spatial task conditions and temporal availability windows are imposed simultaneously [19], [20]. Signal temporal logic (STL) provides a more expressive formalism by combining spatial predicates with temporal operators, and has been integrated with different planning and control methods [21], [22], [23], [24]. However, these approaches do not directly provide a solution to the dwell-time constraint considered in this paper.

To handle the dwell-time constraint more efficiently, this paper uses control barrier functions (CBFs). CBFs enforce safety by rendering a prescribed safe set forward invariant. For control-affine nonlinear systems, the CBF condition is a linear constraint on the control input and can be enforced online via quadratic programming (QP) [25], [26]. However, most existing CBF constructions focus on instantaneous spatial safety relations, such as maintaining separation from static or moving obstacles [27], [28], [29], [30]. Representative examples include collision-cone CBF [31], high-order CBF (HoCBF) [32], [33], CBF based on rotating supporting hyperplanes [34], and CBF based on trajectory circle [35]. Although STL can also be combined with CBFs for temporal specifications [36], [37], [38], the logical predicates and coordination rules required by such formulations may increase both design and computational complexity. While the dwell-time requirement considered here can in principle be represented using STL, their direct incorporation into CBF-based control is nontrivial because the temporal constraint is defined on each continuous traversal of the TLZ: the available dwell-time budget is initialized upon entry, evolves during occupancy, and is reset after exit. Existing STL-CBF formulations may therefore require additional temporal predicates and coordination mechanisms to accommodate such entry-dependent and recurrent timing

behavior. This motivates the direct characterization of the dwell-time constraint and its integration into a CBF formulation without relying on high-level temporal specifications.

This paper proposes a predictive linear trajectory control barrier function (PLTCBF) approach for handling dwell-time constraints in 3D UAV flight. The main contributions are summarized as follows:

- A CBF formulation is developed to enforce dwell-time constraints without relying on high-level temporal logic specifications. By differentiating the proposed PLTCBF along the UAV dynamics, the dwell-time constraint is converted into an affine constraint on the control input and enforced online through a quadratic-programming (QP)-based safety filter. This formulation avoids the additional design complexity associated with logical predicates and temporal operators.
- A PLT is introduced to preview the UAV's future motion from its current state. By combining the PLT with the known motion of the TLZ, a relative PLT is obtained to capture the future relative motion between the UAV and the TLZ. To facilitate geometric analysis, the relative PLT and the TLZ are represented in a transformed space, where ellipsoidal TLZs are mapped to equivalent spheres and the horizontal cross-sections of elliptic-cylinder TLZs are mapped to equivalent circles. This transformed relative geometry yields an explicit characterization of the dwell-time constraint and provides the basis for constructing the PLTCBF.
- The proposed PLTCBF is constructed from three component CBFs for different UAV-TLZ relations. The validity of the individual components and the feasibility of their associated CBF-QPs are established. Safe transitions between the components are proved when the UAV reaches a TLZ boundary or the relative velocity becomes zero. These results guarantee that the dwell-time constraint is satisfied for every TLZ traversal.

The rest of this article is organized as follows. Section II introduces the mathematical preliminaries and formulates the control problem under the dwell-time constraint. Section III presents the PLTCBF design. Section IV establishes the validity, switching safety, and dwell-time guarantee of the PLTCBF. Section V provides numerical simulations to verify the proposed method. Section VI presents a closed-loop validation in CoppeliaSim, and Section VII concludes this article.

II. PROBLEM FORMULATION AND PRELIMINARIES

A. Dynamics Model of the UAV

A UAV in three-dimensional space is modeled as [39]

$$\begin{cases} \dot{\mathbf{p}} = \mathbf{v} \\ \dot{\mathbf{v}} = \mathbf{u} \end{cases} \quad (1)$$

where $\mathbf{p} = [p_x, p_y, p_z]^\top$, $\mathbf{v} = [v_x, v_y, v_z]^\top$, and $\mathbf{u} = [a_x, a_y, a_z]^\top$ denote the position, velocity, and acceleration vectors in $\mathbb{R}^3$, respectively. The state is $\mathbf{x} = [\mathbf{p}^\top, \mathbf{v}^\top]^\top \in \mathbb{R}^6$, with $\mathbf{u}$ as the control input. Moreover, this model can be rewritten in the control-affine form $\dot{\mathbf{x}} = \mathbf{f}(\mathbf{x}) + \mathbf{g}(\mathbf{x})\mathbf{u}$, where $\mathbf{f}(\mathbf{x}) = [\mathbf{v}^\top, \mathbf{0}_{1 \times 3}]^\top$ and $\mathbf{g}(\mathbf{x}) = [\mathbf{0}_{3 \times 3}, \mathbf{I}_3]^\top$.

B. Constraint Description

(1) Dwell-Time Constraint

Let $\mathcal{W} \subset \mathbb{R}^3$ be the workspace and $\mathbf{p} \in \mathcal{W}$ be the UAV position. A TLZ is represented by a time-varying region $\mathcal{T} \subset \mathcal{W}$ whose center moves at constant velocity, while its shape, size, and orientation remain fixed. The dwell interval, dwell duration, and dwell-time constraint are defined below.

Definition 1 (Dwell Interval and Dwell Duration). *Let $\mathcal{T}(t)$ denote the TLZ at time t . For a continuous traversal of the TLZ by the UAV, a time interval $[t_{\text{in}}, t_{\text{out}}]$ is called a dwell interval if $\mathbf{p}(t_{\text{in}}) \in \partial\mathcal{T}(t_{\text{in}})$, $\mathbf{p}(t_{\text{out}}) \in \partial\mathcal{T}(t_{\text{out}})$, and $\mathbf{p}(t) \in \text{Int}(\mathcal{T}(t))$ for all $t \in (t_{\text{in}}, t_{\text{out}})$. Here, t_{in} and t_{out} are the entry and exit times of this traversal, respectively, and the dwell duration is $t_{\text{out}} - t_{\text{in}}$. The notation $\text{Int}(\mathcal{T}(t))$ denotes the interior of the TLZ, excluding its boundary, while $\partial\mathcal{T}(t)$ denotes the boundary of the TLZ.*

Definition 2 (Dwell-Time Constraint). *For each dwell interval $[t_{\text{in}}, t_{\text{out}}]$, the dwell duration must satisfy $t_{\text{out}} - t_{\text{in}} \leq T_{\text{lim}}$, where $T_{\text{lim}} \geq 0$ is the dwell-time limit for the current traversal. At the entry time t_{in} , the remaining dwell-time allowance is initialized as $T_{\text{rem}}(t_{\text{in}}) = T_{\text{lim}}$ and evolves according to $\dot{T}_{\text{rem}} = -1$ while the UAV remains inside the TLZ. At the exit time t_{out} , T_{rem} is no longer updated, and a new dwell-time limit $T_{\text{lim}}^{\text{next}}$ is assigned for a possible subsequent traversal.*

The above definitions are independent of the TLZ shape. For the geometric construction below, we consider two representative TLZ models: ellipsoids and vertical elliptic cylinders.

(a) *Ellipsoidal TLZ*: An ellipsoidal TLZ is centered at $\mathbf{p}_{\text{tlz}} \in \mathbb{R}^3$ and has positive semi-axis lengths $\mathbf{r}_{\text{tlz}} = [r_a, r_b, r_c]^\top$, where $r_a, r_b, r_c > 0$, $r_a \geq r_b$, and $r_a \geq r_c$. Its orientation is described by the ZYX Euler angles $\phi = [\varphi, \theta, \psi]^\top$. The corresponding rotation matrix is

$$\mathbf{R}_3 = \begin{bmatrix} c_\psi c_\theta & c_\psi s_\theta s_\varphi - s_\psi c_\varphi & c_\psi s_\theta c_\varphi + s_\psi s_\varphi \\ s_\psi c_\theta & s_\psi s_\theta s_\varphi + c_\psi c_\varphi & s_\psi s_\theta c_\varphi - c_\psi s_\varphi \\ -s_\theta & c_\theta s_\varphi & c_\theta c_\varphi \end{bmatrix} \quad (2)$$

where $c_\psi = \cos \psi$, $s_\psi = \sin \psi$, $c_\theta = \cos \theta$, $s_\theta = \sin \theta$, $c_\varphi = \cos \varphi$, and $s_\varphi = \sin \varphi$. Let $\mathbf{\Lambda}_3 = \text{diag}(r_a, r_b, r_c)$. The quadratic-form matrix $\mathbf{Q} = \mathbf{R}_3 \mathbf{\Lambda}_3^{-2} \mathbf{R}_3^\top$ then gives the ellipsoidal TLZ as

$$\mathcal{T} = \{\mathbf{w} \in \mathcal{W} \mid (\mathbf{w} - \mathbf{p}_{\text{tlz}})^\top \mathbf{Q} (\mathbf{w} - \mathbf{p}_{\text{tlz}}) \leq 1\} \quad (3)$$

where $\mathbf{w} = [w_x, w_y, w_z]^\top$ is an arbitrary point in the workspace.

(b) *Elliptic-Cylinder TLZ*: Some TLZs are specified primarily by a horizontal footprint, with no explicit constraint imposed on altitude, such as weather-affected or signal-degraded regions. These TLZs are modeled as elliptic cylinders aligned with the z -axis. The cylinder center is $\mathbf{p}_{\text{tlz}} = [p_{\text{tlz},x}, p_{\text{tlz},y}, 0]^\top$, and its horizontal cross-section has positive semi-axis lengths $\mathbf{r}_{\text{tlz}} = [r_a, r_b]^\top$, with $r_a \geq r_b > 0$, and rotation angle ψ . The corresponding horizontal rotation matrix is

$$\mathbf{R}_2 = \begin{bmatrix} \cos \psi & -\sin \psi \\ \sin \psi & \cos \psi \end{bmatrix} \quad (4)$$

Let $\Lambda_2 = \text{diag}(r_a, r_b)$. The matrix $\mathbf{Q}_2 = \mathbf{R}_2 \Lambda_2^{-2} \mathbf{R}_2^\top$ gives the elliptic-cylinder TLZ as

$$\mathcal{T} = \{ \mathbf{w} \in \mathcal{W} \mid (\mathbf{w}_{xy} - \mathbf{p}_{\text{tlz},xy})^\top \mathbf{Q}_2 (\mathbf{w}_{xy} - \mathbf{p}_{\text{tlz},xy}) \leq 1 \} \quad (5)$$

where $\mathbf{w}_{xy} = [w_x, w_y]^\top$ and $\mathbf{p}_{\text{tlz},xy} = [p_{\text{tlz},x}, p_{\text{tlz},y}]^\top$ denote the projections of $\mathbf{w}$ and $\mathbf{p}_{\text{tlz}}$ onto the horizontal plane, respectively.

(2) Control Saturation Constraint

Due to the physical limits of the UAV actuators, the control input $\mathbf{u}$ is subject to the following bounds

$$\mathbf{u}_{\min} \leq \mathbf{u} \leq \mathbf{u}_{\max} \quad (6)$$

where $\mathbf{u}_{\min}, \mathbf{u}_{\max} \in \mathbb{R}^3$ denote the lower and upper acceleration bounds, respectively. The admissible control set is defined as $\mathcal{U} := \{ \mathbf{u} \in \mathbb{R}^3 \mid \mathbf{u}_{\min} \leq \mathbf{u} \leq \mathbf{u}_{\max} \}$. We assume that the zero input lies in the interior of the admissible control set, i.e., $\mathbf{0} \in \text{int } \mathcal{U}$, or equivalently $\mathbf{u}_{\min} < \mathbf{0} < \mathbf{u}_{\max}$ componentwise.

C. Control Barrier Function

Consider the nonlinear control-affine system

$$\dot{\mathbf{x}} = \mathbf{f}(\mathbf{x}) + \mathbf{g}(\mathbf{x})\mathbf{u} \quad (7)$$

where $\mathbf{x} \in \mathcal{D} \subseteq \mathbb{R}^n$ represents the state vector, $\mathbf{u} \in U \subseteq \mathbb{R}^m$ represents the control input, and the functions $\mathbf{f} : \mathbb{R}^n \rightarrow \mathbb{R}^n$, $\mathbf{g} : \mathbb{R}^n \rightarrow \mathbb{R}^{n \times m}$ are locally Lipschitz continuous on $\mathcal{D}$. Let $h : \mathbb{R}^n \rightarrow \mathbb{R}$ be a continuously differentiable function. Then, a set $\mathcal{C}$ is defined as the safe set, which is the superlevel set of h

$$\begin{cases} \mathcal{C} = \{ \mathbf{x} \in \mathcal{D} \subseteq \mathbb{R}^n : h \geq 0 \} \\ \partial\mathcal{C} = \{ \mathbf{x} \in \mathcal{D} \subseteq \mathbb{R}^n : h = 0 \} \\ \text{Int}(\mathcal{C}) = \{ \mathbf{x} \in \mathcal{D} \subseteq \mathbb{R}^n : h > 0 \} \end{cases} \quad (8)$$

Definition 3 (Control Barrier Function [25], [26]). *The function h is a control barrier function (CBF) for the system in Eq. (7) on the safe set $\mathcal{C}$ if there exists an extended class $\mathcal{K}_\infty$ function $\kappa(\cdot)$ such that, for every $\mathbf{x} \in \mathcal{C}$, an admissible input can be chosen to satisfy*

$$\dot{h} + \kappa(h) \geq 0 \quad (9)$$

where $\dot{h}$ denotes the total derivative of the CBF along the system in Eq. (7) under the chosen input.

For $h = h(\mathbf{x})$, this condition becomes

$$\sup_{\mathbf{u} \in U} [\mathcal{L}_f h(\mathbf{x}) + \mathcal{L}_g h(\mathbf{x})\mathbf{u} + \kappa(h(\mathbf{x}))] \geq 0 \quad (10)$$

where $\mathcal{L}_f h(\mathbf{x})$ and $\mathcal{L}_g h(\mathbf{x})$ represent the Lie derivatives of $h(\mathbf{x})$ along $\mathbf{f}$ and $\mathbf{g}$, respectively.

If the CBF explicitly depends on time, i.e., $h = h(\mathbf{x}, t)$, the condition in Eq. (9) becomes

$$\sup_{\mathbf{u} \in U} [\partial_t h(\mathbf{x}, t) + \mathcal{L}_f h(\mathbf{x}, t) + \mathcal{L}_g h(\mathbf{x}, t)\mathbf{u} + \kappa(h(\mathbf{x}, t))] \geq 0 \quad (11)$$

where $\partial_t h(\mathbf{x}, t) = \partial h(\mathbf{x}, t) / \partial t$.

Remark 1. The choice of the extended class $\mathcal{K}_\infty$ function κ influences the system behavior near the boundary of the safe set $\mathcal{C}$. A common choice is $\kappa(h) = kh$, with $k > 0$. The Lie derivative $\mathcal{L}_g h$ quantifies the influence of the control



Fig. 1. CBF-QP safety filter architecture

input on the evolution of h . If $\mathcal{L}_g h \neq \mathbf{0}$, the CBF conditions in Eqs. (10) and (11) depend explicitly on $\mathbf{u}$ and therefore impose constraints on the control input.

D. CBF-QP Formulation

To enforce safety, the CBF condition is imposed as a constraint in a quadratic programming (QP)-based safety filter that minimally modifies the nominal control input. Equation (10) applies to $h = h(\mathbf{x})$, whereas Eq. (11) applies to $h = h(\mathbf{x}, t)$. Both conditions are affine in $\mathbf{u}$ and can be expressed in the unified form $A(\mathbf{x}, t)\mathbf{u} \leq b(\mathbf{x}, t)$. Here, $A(\mathbf{x}, t) = -\mathcal{L}_g h(\mathbf{x}, t)$ and $b(\mathbf{x}, t) = \partial_t h(\mathbf{x}, t) + \mathcal{L}_f h(\mathbf{x}, t) + \kappa(h(\mathbf{x}, t))$, where $\partial_t h(\mathbf{x}, t) = 0$ for $h = h(\mathbf{x})$. Given a nominal control input $\mathbf{u}_{\text{nom}}$, the safety-filtered command $\mathbf{u}_{\text{safety}}$ is obtained by solving the following CBF-QP:

Definition 4 (CBF-QP [25], [26]).

$$\begin{aligned} \mathbf{u}_{\text{safety}} &= \arg \min_{\mathbf{u}} \|\mathbf{u} - \mathbf{u}_{\text{nom}}\|^2 \\ \text{s.t. } & A(\mathbf{x}, t)\mathbf{u} \leq b(\mathbf{x}, t) \\ & \mathbf{u}_{\min} \leq \mathbf{u} \leq \mathbf{u}_{\max} \end{aligned} \quad (12)$$

The corresponding architecture is shown in Fig. 1.

III. CONSTRUCTION OF PLTCBF

A. Predictive Linear Trajectory (PLT)

Given the current UAV state $(\mathbf{p}, \mathbf{v})$, the velocity is instantaneously assumed constant to construct a straight-line geometric predictor. The resulting trajectory $\mathbf{p}(t + \tau) = \mathbf{p} + \mathbf{v}\tau$, $\tau \geq 0$, is referred to as the predictive linear trajectory (PLT).

The TLZ center may also move with known constant velocity $\mathbf{v}_{\text{tlz}}$, so $\mathbf{p}_{\text{tlz}}(t + \tau) = \mathbf{p}_{\text{tlz}} + \mathbf{v}_{\text{tlz}}\tau$. Combining this motion with the UAV PLT gives the relative PLT $\mathbf{p}_{\text{rel}}(t + \tau) = \mathbf{p}_{\text{rel}} + \mathbf{v}_{\text{rel}}\tau$, where $\mathbf{p}_{\text{rel}} = \mathbf{p} - \mathbf{p}_{\text{tlz}}$ and $\mathbf{v}_{\text{rel}} = \mathbf{v} - \mathbf{v}_{\text{tlz}}$. It describes the UAV's predicted motion with respect to the TLZ.

B. Coordinate Transformation of the TLZ

To facilitate geometric analysis, the relative PLT and the TLZ are represented in a transformed space. In this space, an ellipsoidal TLZ is mapped to an equivalent sphere, whereas the horizontal cross-section of an elliptic-cylinder TLZ is mapped to an equivalent circle.

(1) Transformation of Ellipsoidal TLZ

For an ellipsoidal TLZ, $\mathbf{R}_3^\top$ rotates the world frame into its principal-axis frame, and $\mathbf{S}_3 = \text{diag}(1, r_a/r_b, r_a/r_c)$ rescales the second and third axes to r_a . The resulting transformation matrix is $\mathbf{T}_3 = \mathbf{S}_3 \mathbf{R}_3^\top$. With $\tilde{\mathbf{w}} = \mathbf{T}_3 \mathbf{w}$ and $\tilde{\mathbf{p}}_{\text{tlz}} = \mathbf{T}_3 \mathbf{p}_{\text{tlz}}$, the ellipsoidal TLZ is mapped to a sphere of radius $\tilde{r}_{\text{tlz}} = r_a$, i.e., $\|\tilde{\mathbf{w}} - \tilde{\mathbf{p}}_{\text{tlz}}\| \leq \tilde{r}_{\text{tlz}}$.

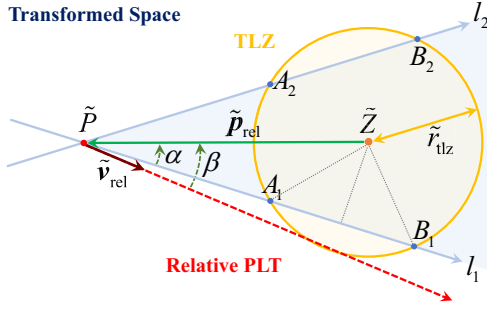


Fig. 2. Geometric illustration of the relative PLT and the TLZ in the transformed space

(2) Transformation of Elliptic-Cylinder TLZ

For an elliptic-cylinder TLZ, the transformation is applied to its horizontal projection. The transformation matrix $T_2 = S_2 R_2^\top$, with $S_2 = \text{diag}(1, r_a/r_b)$, rotates the elliptic cross-section into its principal-axis frame and rescales its minor axis to r_a , thereby mapping it to a disk of radius $\tilde{r}_{\text{tlz}} = r_a$.

(3) Transformed Relative Motion

A single notation is used for both TLZ types. Let $d = 3$ for an ellipsoidal TLZ and $d = 2$ for an elliptic-cylinder TLZ. Define $E_3 = I_3$ and $E_2 = [I_2 \ 0_{2 \times 1}]$, where E_d selects the coordinates used in the transformation. Set $T_d = T_3$ for $d = 3$ and $T_d = T_2$ for $d = 2$. Since these transformations are linear, applying them to the relative position and velocity gives

$$\tilde{p}_{\text{rel}} = T_d E_d p_{\text{rel}}, \quad \tilde{v}_{\text{rel}} = T_d E_d v_{\text{rel}} \quad (13)$$

where $\tilde{p}_{\text{rel}}, \tilde{v}_{\text{rel}} \in \mathbb{R}^d$.

Lemma 1. For $d \in \{2, 3\}$ and any $a \in \mathbb{R}^d$, we have

$$a^\top T_d E_d = 0 \implies a = 0 \quad (14)$$

Proof. The matrix E_d has full row rank and satisfies $E_d E_d^\top = I_d$. Therefore

$$a^\top T_d E_d = 0 \implies a^\top T_d = a^\top T_d E_d E_d^\top = 0 \quad (15)$$

Since T_d is nonsingular in $\mathbb{R}^d$, it follows that $a = 0$. $\square$

C. PLTCBF Design

The PLTCBF is constructed in the transformed d -dimensional space, where the TLZ is represented by the unified condition $\|\tilde{p}_{\text{rel}}\| \leq \tilde{r}_{\text{tlz}}$. Here, $d = 3$ for ellipsoidal TLZs and $d = 2$ for elliptic-cylinder TLZs, whose horizontal cross-sections are considered. As illustrated in Fig. 2, $\tilde{P}$ and $\tilde{Z}$ denote the UAV position and TLZ center, respectively, and $\tilde{p}_{\text{rel}}$ points from $\tilde{Z}$ to $\tilde{P}$. The relative PLT is the ray from $\tilde{P}$ along $\tilde{v}_{\text{rel}}$.

Based on the relationship between the UAV and the TLZ, we consider the following cases.

Case I: Outside the TLZ

For $\|\tilde{p}_{\text{rel}}\| > \tilde{r}_{\text{tlz}}$, unsafe PLT directions are those whose chord length inside the TLZ exceeds $\|\tilde{v}_{\text{rel}}\| T_{\text{lim}}$. In the nontrivial case $\|\tilde{v}_{\text{rel}}\| T_{\text{lim}} \leq 2\tilde{r}_{\text{tlz}}$, the boundary directions define a cone, referred to here as the critical cone, as shown

in Fig. 2. Its axis is $\tilde{P}\tilde{Z}$, whose direction is $-\tilde{p}_{\text{rel}}$, and its half-angle α satisfies

$$\cos \alpha = \frac{\sqrt{\|\tilde{p}_{\text{rel}}\|^2 - \tilde{r}_{\text{tlz}}^2 + \frac{\|\tilde{v}_{\text{rel}}\|^2 T_{\text{lim}}^2}{4}}}{\|\tilde{p}_{\text{rel}}\|} \quad (16)$$

Let β be the angle between the PLT direction $\tilde{v}_{\text{rel}}$ and the cone-axis direction $-\tilde{p}_{\text{rel}}$. Then, we have

$$\cos \beta = \frac{-\tilde{p}_{\text{rel}}^\top \tilde{v}_{\text{rel}}}{\|\tilde{p}_{\text{rel}}\| \|\tilde{v}_{\text{rel}}\|} \quad (17)$$

For $\|\tilde{v}_{\text{rel}}\| > 0$, the dwell-time constraint requires $\beta \geq \alpha$, or equivalently $\cos \alpha \geq \cos \beta$. Using Eqs. (16) and (17) gives

$$\tilde{p}_{\text{rel}}^\top \tilde{v}_{\text{rel}} + \|\tilde{v}_{\text{rel}}\| \sqrt{\|\tilde{p}_{\text{rel}}\|^2 - \tilde{r}_{\text{tlz}}^2 + \frac{\|\tilde{v}_{\text{rel}}\|^2 T_{\text{lim}}^2}{4}} \geq 0 \quad (18)$$

This inequality motivates the following candidate CBF

$$h_1(x) = \tilde{p}_{\text{rel}}^\top \tilde{v}_{\text{rel}} + \|\tilde{v}_{\text{rel}}\| \sqrt{q} \quad (19)$$

where q is defined as

$$q = \|\tilde{p}_{\text{rel}}\|^2 - \tilde{r}_{\text{tlz}}^2 + \frac{\|\tilde{v}_{\text{rel}}\|^2 T_{\text{lim}}^2}{4} \quad (20)$$

For Case I with $\|\tilde{v}_{\text{rel}}\| > 0$, $q > 0$ follows from $\|\tilde{p}_{\text{rel}}\| > \tilde{r}_{\text{tlz}}$, so $h_1(x)$ is well defined. The geometric derivation above uses a real half-angle α . If $\|\tilde{v}_{\text{rel}}\| T_{\text{lim}} > 2\tilde{r}_{\text{tlz}}$, the right-hand side of Eq. (16) is larger than one, so no real critical-cone boundary exists. In this case, since $\sqrt{q} > \|\tilde{p}_{\text{rel}}\|$ and $\tilde{p}_{\text{rel}}^\top \tilde{v}_{\text{rel}} \geq -\|\tilde{p}_{\text{rel}}\| \|\tilde{v}_{\text{rel}}\|$, we have $h_1(x) > 0$, and the dwell-time constraint is automatically satisfied by the PLT. At the TLZ boundary, $h_1(x)$ is smooth for $T_{\text{lim}} > 0$; the $T_{\text{lim}} = 0$ case is addressed in the subsequent analysis.

Case II: Inside or entering the TLZ

This case covers $\|\tilde{p}_{\text{rel}}\| < \tilde{r}_{\text{tlz}}$ and boundary states at an entry event, defined as a crossing from the exterior into the interior of the TLZ. Such a boundary event satisfies $\|\tilde{p}_{\text{rel}}\| = \tilde{r}_{\text{tlz}}$ and $\tilde{p}_{\text{rel}}^\top \tilde{v}_{\text{rel}} \leq 0$. At entry, set $T_{\text{rem}} = T_{\text{lim}}$. The current relative PLT predicts the relative position after T_{rem} as

$$s = \tilde{p}_{\text{rel}} + \tilde{v}_{\text{rel}} T_{\text{rem}} \quad (21)$$

Requiring the predicted relative position s to lie on or outside the TLZ boundary gives $\|s\| \geq \tilde{r}_{\text{tlz}}$, which motivates the following candidate CBF

$$h_2(x, t) = s^\top s - \tilde{r}_{\text{tlz}}^2 \quad (22)$$

Case III: Auxiliary exterior mode

This case handles exterior states at which $h_1(x)$ cannot be enforced. If $\|\tilde{v}_{\text{rel}}\| = 0$, the expression in Eq. (19) is continuous but not differentiable because of the velocity norm. Moreover, when $T_{\text{lim}} = 0$ and $\|\tilde{p}_{\text{rel}}\| = \tilde{r}_{\text{tlz}}$, we have $q = 0$, so $h_1(x)$ is also not differentiable on the TLZ boundary. These cases are handled by the candidate CBF

$$h_3(x) = \tilde{p}_{\text{rel}}^\top \tilde{v}_{\text{rel}} \quad (23)$$

For the subsequent analysis, define $\mathcal{D}_{10} = \{x : \|\tilde{p}_{\text{rel}}\| \geq \tilde{r}_{\text{tlz}}\}$, $\mathcal{D}_b = \{x : T_{\text{lim}} = 0, \|\tilde{p}_{\text{rel}}\| = \tilde{r}_{\text{tlz}}\}$, $\mathcal{D}_1 = \{x \in \mathcal{D}_{10} \setminus \mathcal{D}_b : \|\tilde{v}_{\text{rel}}\| > 0\}$, $\mathcal{D}_2 = \{x : \|\tilde{p}_{\text{rel}}\| \leq \tilde{r}_{\text{tlz}}\}$, and $\mathcal{D}_3 = \mathcal{D}_{10}$.

The functions $h_1(\mathbf{x})$, $h_2(\mathbf{x}, t)$, and $h_3(\mathbf{x})$ are referred to as the component CBFs that constitute the PLTCBF. Let $i_{\text{act}}(t) \in \{1, 2, 3\}$ denote the active mode, which selects the component CBF enforced at time t . The resulting PLTCBF is defined as

$$h_{i_{\text{act}}}(\mathbf{x}, t) = \begin{cases} h_1(\mathbf{x}), & i_{\text{act}}(t) = 1 \\ h_2(\mathbf{x}, t), & i_{\text{act}}(t) = 2 \\ h_3(\mathbf{x}), & i_{\text{act}}(t) = 3 \end{cases} \quad (24)$$

The active mode $i_{\text{act}}(t)$ is determined by the UAV-TLZ relation. Outside the TLZ, $i_{\text{act}} = 1$ when $\|\tilde{\mathbf{v}}_{\text{rel}}\| > 0$, whereas $i_{\text{act}} = 3$ when $\|\tilde{\mathbf{v}}_{\text{rel}}\| = 0$. For $T_{\text{lim}} > 0$, the switch to $i_{\text{act}} = 2$ occurs when an entry event from the exterior into the interior is detected with $\|\tilde{\mathbf{v}}_{\text{rel}}\| > 0$; such an event satisfies $\tilde{\mathbf{p}}_{\text{rel}}^\top \tilde{\mathbf{v}}_{\text{rel}} \leq 0$, and T_{rem} is initialized at this switch. If $\|\tilde{\mathbf{v}}_{\text{rel}}\| = 0$ on the boundary, $i_{\text{act}} = 3$ is selected instead, and enforcing $h_3(\mathbf{x}) \geq 0$ prevents an inward crossing. At the exit boundary, $h_2(\mathbf{x}, t)$ is deactivated and the next dwell-time limit $T_{\text{lim}}^{\text{next}}$ is loaded. If $\|\tilde{\mathbf{v}}_{\text{rel}}\| > 0$, the active mode switches to 1 when $T_{\text{lim}}^{\text{next}} > 0$ and to 3 when $T_{\text{lim}}^{\text{next}} = 0$; in the latter case, it switches to 1 once the UAV departs from the boundary, where $h_1(\mathbf{x})$ becomes differentiable. If $\|\tilde{\mathbf{v}}_{\text{rel}}\| = 0$ at the exit boundary, $i_{\text{act}} = 3$ is selected and switches to 1 once the UAV departs from the boundary with $\|\tilde{\mathbf{v}}_{\text{rel}}\| > 0$.

Remark 2. The PLT should not be interpreted as a fixed open-loop prediction that the UAV is assumed to follow over the entire remaining dwell-time interval. Instead, it is an instantaneous geometric predictor constructed from the current closed-loop state. At each sampling instant, the PLT is recomputed from the updated state to formulate the active PLTCBF constraint, which is then enforced in the CBF-QP. Moreover, $h_1(\mathbf{x})$, $h_2(\mathbf{x}, t)$, and $h_3(\mathbf{x})$ retain relative degree one, meaning that the control input appears in their first time derivatives. Consequently, the safety constraints can be imposed directly in the CBF-QP without higher-order CBF construction or its associated parameter selection. The geometric construction also avoids online reachability analysis and forward simulation.

IV. ANALYSIS OF PLTCBF

A. Validity of PLTCBF

We establish the validity of the three component CBFs that constitute the PLTCBF. We begin with $h_1(\mathbf{x})$ by defining its associated safe set and zero-level set as $\mathcal{C}_1 = \{\mathbf{x} \in \mathcal{D}_1 : h_1(\mathbf{x}) \geq 0\}$ and $\partial\mathcal{C}_1 = \{\mathbf{x} \in \mathcal{D}_1 : h_1(\mathbf{x}) = 0\}$, respectively.

Using $\tilde{\mathbf{p}}_{\text{rel}} = \mathbf{T}_d \mathbf{E}_d \mathbf{p}_{\text{rel}}$ and $\tilde{\mathbf{v}}_{\text{rel}} = \mathbf{T}_d \mathbf{E}_d \mathbf{v}_{\text{rel}}$, we have $\dot{\tilde{\mathbf{p}}}_{\text{rel}} = \tilde{\mathbf{v}}_{\text{rel}}$ and $\dot{\tilde{\mathbf{v}}}_{\text{rel}} = \mathbf{T}_d \mathbf{E}_d \mathbf{u}$. On $\mathcal{D}_1$, the Lie derivatives of $h_1(\mathbf{x})$ are

$$\mathcal{L}_f h_1(\mathbf{x}) = \|\tilde{\mathbf{v}}_{\text{rel}}\|^2 + \|\tilde{\mathbf{v}}_{\text{rel}}\| \frac{\tilde{\mathbf{p}}_{\text{rel}}^\top \tilde{\mathbf{v}}_{\text{rel}}}{\sqrt{q}} \quad (25)$$

$$\mathcal{L}_g h_1(\mathbf{x}) = \left[\tilde{\mathbf{p}}_{\text{rel}} + \left(\frac{\sqrt{q}}{\|\tilde{\mathbf{v}}_{\text{rel}}\|} + \frac{\|\tilde{\mathbf{v}}_{\text{rel}}\| T_{\text{lim}}^2}{4\sqrt{q}} \right) \tilde{\mathbf{v}}_{\text{rel}} \right]^\top \mathbf{T}_d \mathbf{E}_d \quad (26)$$

Theorem 1 (Validity of h_1). $h_1(\mathbf{x})$ is continuously differentiable on $\mathcal{D}_1$ and satisfies $\mathcal{L}_g h_1(\mathbf{x}) \neq \mathbf{0}$ for all $\mathbf{x} \in \mathcal{C}_1$.

Moreover, for every $\mathbf{x} \in \mathcal{C}_1$, the CBF-QP in Eq. (12) associated with $h_1(\mathbf{x})$ is feasible. Hence, $h_1(\mathbf{x})$ is a valid CBF on $\mathcal{C}_1$.

Proof. For any $\mathbf{x} \in \mathcal{D}_1$, either $\|\tilde{\mathbf{p}}_{\text{rel}}\| > \tilde{r}_{\text{tlz}}$ or $\|\tilde{\mathbf{p}}_{\text{rel}}\| = \tilde{r}_{\text{tlz}}$ with $T_{\text{lim}} > 0$. In the first situation, Eq. (20) directly gives $q > 0$; in the second, it gives $q = \frac{1}{4} \|\tilde{\mathbf{v}}_{\text{rel}}\|^2 T_{\text{lim}}^2 > 0$. Hence, $q > 0$ on $\mathcal{D}_1$. Therefore, $\sqrt{q}$ and $\|\tilde{\mathbf{v}}_{\text{rel}}\|$ are continuously differentiable, and so is $h_1(\mathbf{x})$.

It remains to prove that $\mathcal{L}_g h_1(\mathbf{x}) \neq \mathbf{0}$ on $\mathcal{C}_1$. Suppose that $\mathcal{L}_g h_1(\mathbf{x}) = \mathbf{0}$ for some $\mathbf{x} \in \mathcal{C}_1$. By Lemma 1, Eq. (26) then implies

$$\tilde{\mathbf{p}}_{\text{rel}} + \gamma \tilde{\mathbf{v}}_{\text{rel}} = \mathbf{0} \quad (27)$$

where $\gamma = \frac{\sqrt{q}}{\|\tilde{\mathbf{v}}_{\text{rel}}\|} + \frac{\|\tilde{\mathbf{v}}_{\text{rel}}\| T_{\text{lim}}^2}{4\sqrt{q}}$.

Taking the inner product of Eq. (27) with $\tilde{\mathbf{v}}_{\text{rel}}$ gives $\tilde{\mathbf{p}}_{\text{rel}}^\top \tilde{\mathbf{v}}_{\text{rel}} = -\gamma \|\tilde{\mathbf{v}}_{\text{rel}}\|^2$. Substituting this relation into Eq. (19) yields

$$h_1(\mathbf{x}) = \|\tilde{\mathbf{v}}_{\text{rel}}\| (\sqrt{q} - \gamma \|\tilde{\mathbf{v}}_{\text{rel}}\|) = -\frac{\|\tilde{\mathbf{v}}_{\text{rel}}\|^3 T_{\text{lim}}^2}{4\sqrt{q}} \quad (28)$$

If $T_{\text{lim}} > 0$, then $h_1(\mathbf{x}) < 0$, which contradicts $\mathbf{x} \in \mathcal{C}_1$. Therefore, $\mathcal{L}_g h_1(\mathbf{x}) = \mathbf{0}$ in $\mathcal{C}_1$ could only occur when $T_{\text{lim}} = 0$. For $T_{\text{lim}} = 0$, we have $\gamma = \sqrt{q}/\|\tilde{\mathbf{v}}_{\text{rel}}\|$. Then Eq. (27) gives $\|\tilde{\mathbf{p}}_{\text{rel}}\| = \sqrt{q}$. However, Eq. (20) reduces to $q = \|\tilde{\mathbf{p}}_{\text{rel}}\|^2 - \tilde{r}_{\text{tlz}}^2$, which would imply $\tilde{r}_{\text{tlz}} = 0$, contradicting $\tilde{r}_{\text{tlz}} > 0$. Thus, $\mathcal{L}_g h_1(\mathbf{x}) \neq \mathbf{0}$ on $\mathcal{C}_1$.

It remains to verify the feasibility of the corresponding CBF-QP. For any $\mathbf{x} \in \mathcal{C}_1$, $h_1(\mathbf{x}) \geq 0$ implies $\tilde{\mathbf{p}}_{\text{rel}}^\top \tilde{\mathbf{v}}_{\text{rel}} \geq -\|\tilde{\mathbf{v}}_{\text{rel}}\| \sqrt{q}$. Together with Eq. (25), this gives $\mathcal{L}_f h_1(\mathbf{x}) \geq 0$. Choosing $\mathbf{u} = \mathbf{0}$ yields $\mathcal{L}_f h_1(\mathbf{x}) + \mathcal{L}_g h_1(\mathbf{x}) \mathbf{u} + \kappa(h_1(\mathbf{x})) \geq 0$. Since $\mathbf{0} \in \mathcal{U}$, the input bounds in Eq. (12) are also satisfied. Therefore, $\mathbf{u} = \mathbf{0}$ is a feasible solution of the CBF-QP in Eq. (12) associated with $h_1(\mathbf{x})$, and $h_1(\mathbf{x})$ is a valid CBF on $\mathcal{C}_1$. $\square$

We next consider $h_2(\mathbf{x}, t)$, whose associated safe set and zero-level set are $\mathcal{C}_2 = \{\mathbf{x} \in \mathcal{D}_2 : h_2(\mathbf{x}, t) \geq 0\}$ and $\partial\mathcal{C}_2 = \{\mathbf{x} \in \mathcal{D}_2 : h_2(\mathbf{x}, t) = 0\}$, respectively.

Then, using $\mathbf{s} = \tilde{\mathbf{p}}_{\text{rel}} + \tilde{\mathbf{v}}_{\text{rel}} T_{\text{rem}}$ and $\dot{T}_{\text{rem}} = -1$, the derivatives of $h_2(\mathbf{x}, t)$ are

$$\partial_t h_2(\mathbf{x}, t) = -2\mathbf{s}^\top \tilde{\mathbf{v}}_{\text{rel}} \quad (29)$$

$$\mathcal{L}_f h_2(\mathbf{x}, t) = 2\mathbf{s}^\top \tilde{\mathbf{v}}_{\text{rel}} \quad (30)$$

$$\mathcal{L}_g h_2(\mathbf{x}, t) = 2T_{\text{rem}} \mathbf{s}^\top \mathbf{T}_d \mathbf{E}_d \quad (31)$$

Theorem 2 (Validity of h_2). For $T_{\text{rem}} > 0$, $h_2(\mathbf{x}, t)$ is continuously differentiable on $\mathcal{D}_2$ and satisfies $\mathcal{L}_g h_2(\mathbf{x}, t) \neq \mathbf{0}$ for all $\mathbf{x} \in \mathcal{C}_2$. Moreover, for every $\mathbf{x} \in \mathcal{C}_2$, the CBF-QP in Eq. (12) associated with $h_2(\mathbf{x}, t)$ is feasible. Hence, $h_2(\mathbf{x}, t)$ is a valid CBF on $\mathcal{C}_2$.

Proof. For $T_{\text{rem}} > 0$, $\mathbf{s} = \tilde{\mathbf{p}}_{\text{rel}} + \tilde{\mathbf{v}}_{\text{rel}} T_{\text{rem}}$ is smooth in $\tilde{\mathbf{p}}_{\text{rel}}$, $\tilde{\mathbf{v}}_{\text{rel}}$, and T_{rem} . Therefore, $h_2(\mathbf{x}, t) = \mathbf{s}^\top \mathbf{s} - \tilde{r}_{\text{tlz}}^2$ is continuously differentiable on $\mathcal{D}_2$.

For any $\mathbf{x} \in \mathcal{C}_2$, the condition $h_2(\mathbf{x}, t) \geq 0$ implies $\|\mathbf{s}\| \geq \tilde{r}_{\text{tlz}} > 0$, and hence $\mathbf{s} \neq \mathbf{0}$. Suppose, for contradiction, that $\mathcal{L}_g h_2(\mathbf{x}, t) = \mathbf{0}$. From Eq. (31), we have

$$T_{\text{rem}} \mathbf{s}^\top \mathbf{T}_d \mathbf{E}_d = \mathbf{0} \quad (32)$$

Since $T_{\text{rem}} > 0$, Eq. (32) implies $\mathbf{s}^\top \mathbf{T}_d \mathbf{E}_d = \mathbf{0}$. By Lemma 1, this gives $\mathbf{s} = \mathbf{0}$, which contradicts $\|\mathbf{s}\| \geq \tilde{r}_{\text{tlz}} > 0$. Hence, $\mathcal{L}_g h_2(\mathbf{x}, t) \neq \mathbf{0}$ on $\mathcal{C}_2$.

It remains to verify the feasibility of the corresponding CBF-QP. For any $\mathbf{x} \in \mathcal{C}_2$, we have $h_2(\mathbf{x}, t) \geq 0$. Moreover, $\partial_t h_2(\mathbf{x}, t) + \mathcal{L}_f h_2(\mathbf{x}, t) = 0$. Choosing $\mathbf{u} = \mathbf{0}$ gives $\partial_t h_2(\mathbf{x}, t) + \mathcal{L}_f h_2(\mathbf{x}, t) + \mathcal{L}_g h_2(\mathbf{x}, t) \mathbf{u} + \kappa(h_2(\mathbf{x}, t)) = \kappa(h_2(\mathbf{x}, t)) \geq 0$. Since $\mathbf{0} \in \mathcal{U}$, the input bounds in Eq. (12) are also satisfied. Therefore, $\mathbf{u} = \mathbf{0}$ is a feasible solution of the CBF-QP in Eq. (12) associated with $h_2(\mathbf{x}, t)$, and $h_2(\mathbf{x}, t)$ is a valid CBF on $\mathcal{C}_2$. $\square$

Finally, we consider $h_3(\mathbf{x})$. Its associated safe set and zero-level set are $\mathcal{C}_3 = \{\mathbf{x} \in \mathcal{D}_3 : h_3(\mathbf{x}) \geq 0\}$ and $\partial \mathcal{C}_3 = \{\mathbf{x} \in \mathcal{D}_3 : h_3(\mathbf{x}) = 0\}$, respectively.

Using $\dot{\tilde{\mathbf{p}}}_{\text{rel}} = \tilde{\mathbf{v}}_{\text{rel}}$ and $\dot{\tilde{\mathbf{v}}}_{\text{rel}} = \mathbf{T}_d \mathbf{E}_d \mathbf{u}$, the Lie derivatives of $h_3(\mathbf{x})$ are

$$\mathcal{L}_f h_3(\mathbf{x}) = \|\tilde{\mathbf{v}}_{\text{rel}}\|^2 \quad (33)$$

$$\mathcal{L}_g h_3(\mathbf{x}) = \tilde{\mathbf{p}}_{\text{rel}}^\top \mathbf{T}_d \mathbf{E}_d \quad (34)$$

Theorem 3 (Validity of h_3). $h_3(\mathbf{x})$ is continuously differentiable on $\mathcal{D}_3$ and satisfies $\mathcal{L}_g h_3(\mathbf{x}) \neq \mathbf{0}$ for all $\mathbf{x} \in \mathcal{D}_3$. Moreover, for every $\mathbf{x} \in \mathcal{C}_3$, the CBF-QP in Eq. (12) associated with $h_3(\mathbf{x})$ is feasible. Hence, $h_3(\mathbf{x})$ is a valid CBF on $\mathcal{D}_3$.

Proof. The function $h_3(\mathbf{x}) = \tilde{\mathbf{p}}_{\text{rel}}^\top \tilde{\mathbf{v}}_{\text{rel}}$ is smooth in $\tilde{\mathbf{p}}_{\text{rel}}$ and $\tilde{\mathbf{v}}_{\text{rel}}$, hence it is continuously differentiable on $\mathcal{D}_3$.

For any $\mathbf{x} \in \mathcal{D}_3$, $\|\tilde{\mathbf{p}}_{\text{rel}}\| \geq \tilde{r}_{\text{tlz}} > 0$, so $\tilde{\mathbf{p}}_{\text{rel}} \neq \mathbf{0}$. If $\mathcal{L}_g h_3(\mathbf{x}) = \mathbf{0}$, then Eq. (34) gives $\tilde{\mathbf{p}}_{\text{rel}}^\top \mathbf{T}_d \mathbf{E}_d = \mathbf{0}$. By Lemma 1, this would imply $\tilde{\mathbf{p}}_{\text{rel}} = \mathbf{0}$, contradicting $\|\tilde{\mathbf{p}}_{\text{rel}}\| \geq \tilde{r}_{\text{tlz}} > 0$. Hence, $\mathcal{L}_g h_3(\mathbf{x}) \neq \mathbf{0}$ on $\mathcal{D}_3$.

It remains to verify the feasibility of the corresponding CBF-QP. For any $\mathbf{x} \in \mathcal{C}_3$, $h_3(\mathbf{x}) \geq 0$ and $\mathcal{L}_f h_3(\mathbf{x}) = \|\tilde{\mathbf{v}}_{\text{rel}}\|^2 \geq 0$. Since $\mathbf{0} \in \mathcal{U}$, choosing $\mathbf{u} = \mathbf{0}$ satisfies the CBF condition and the input bounds in Eq. (12). Therefore, the CBF-QP associated with $h_3(\mathbf{x})$ is feasible.

When $\tilde{\mathbf{v}}_{\text{rel}} = \mathbf{0}$, the CBF condition reduces to $\mathcal{L}_g h_3(\mathbf{x}) \mathbf{u} \geq 0$. In addition to the feasible zero input, $\mathbf{0} \in \text{int } \mathcal{U}$ guarantees that $\mathbf{u} = \varepsilon [\mathcal{L}_g h_3(\mathbf{x})]^\top \in \mathcal{U}$ for sufficiently small $\varepsilon > 0$, and this input satisfies $\mathcal{L}_g h_3(\mathbf{x}) \mathbf{u} = \varepsilon \|\mathcal{L}_g h_3(\mathbf{x})\|^2 > 0$. The feasibility of $\mathbf{u} = \mathbf{0}$ does not imply that it is always selected by the CBF-QP. For a single CBF constraint associated with $h_3(\mathbf{x})$, the zero solution occurs only when $\mathbf{u}_{\text{nom}} = -\lambda [\mathcal{L}_g h_3(\mathbf{x})]^\top$ for some $\lambda \geq 0$. Since the nominal control generally depends on the current UAV state and task reference, this alignment generally changes as the UAV moves. Even if the zero solution repeats, it gives $\dot{h}_3(\mathbf{x}) = 0$ and therefore preserves safety. Hence, $h_3(\mathbf{x})$ is a valid CBF on $\mathcal{D}_3$. $\square$

B. Switching Safety

Each component CBF preserves its associated safe set while it is active. Because the continuous state is unchanged at a mode switch, switching safety follows if the switching state belongs to the safe set of the newly activated component. We verify this condition for entry switches, exit switches, and switches at states where the relative velocity is zero, beginning with the entry switch from $h_1(\mathbf{x})$ to $h_2(\mathbf{x}, t)$.

Theorem 4 (Entry-boundary switching from h_1 to h_2). Suppose that $T_{\text{lim}} > 0$. Let t_s be an entry event at which the UAV crosses the boundary $\|\tilde{\mathbf{p}}_{\text{rel}}(t_s)\| = \tilde{r}_{\text{tlz}}$ from the exterior into the interior with $\|\tilde{\mathbf{v}}_{\text{rel}}(t_s)\| > 0$. If $h_1(\mathbf{x}) \geq 0$ is maintained before the switch, then the switch from $h_1(\mathbf{x})$ to $h_2(\mathbf{x}, t)$ yields $h_2(\mathbf{x}(t_s), t_s) \geq 0$.

Proof. At the entry boundary, $h_1(\mathbf{x})$ is deactivated and $h_2(\mathbf{x}, t)$ becomes active. Since $T_{\text{lim}} > 0$, $h_1(\mathbf{x}(t))$ is continuous at t_s . Therefore, the maintained condition $h_1(\mathbf{x}(t)) \geq 0$ for $t < t_s$ gives $h_1(\mathbf{x}(t_s)) \geq 0$. At t_s , $T_{\text{rem}}(t_s) = T_{\text{lim}}$ and $q(t_s) = \|\tilde{\mathbf{v}}_{\text{rel}}(t_s)\|^2 T_{\text{lim}}^2 / 4$. Hence, $h_1(\mathbf{x}(t_s)) = \tilde{\mathbf{p}}_{\text{rel}}(t_s)^\top \tilde{\mathbf{v}}_{\text{rel}}(t_s) + \|\tilde{\mathbf{v}}_{\text{rel}}(t_s)\|^2 T_{\text{lim}} / 2$. Substituting $T_{\text{rem}}(t_s) = T_{\text{lim}}$ into $h_2(\mathbf{x}, t)$ gives $h_2(\mathbf{x}(t_s), t_s) = 2T_{\text{lim}} h_1(\mathbf{x}(t_s)) \geq 0$. $\square$

When the UAV reaches the TLZ exit boundary, $h_2(\mathbf{x}, t)$ is deactivated, and a new dwell-time limit $T_{\text{lim}}^{\text{next}}$ is assigned for a possible subsequent traversal. The active component switches to $h_1(\mathbf{x})$ if $T_{\text{lim}}^{\text{next}} > 0$ and $\|\tilde{\mathbf{v}}_{\text{rel}}\| > 0$. If $T_{\text{lim}}^{\text{next}} = 0$ or $\|\tilde{\mathbf{v}}_{\text{rel}}\| = 0$, $h_3(\mathbf{x})$ is selected instead. The following result establishes the safety of both exit-boundary switching branches.

Theorem 5 (Safety of exit-boundary switching from h_2). Suppose that the UAV reaches the exit boundary $\|\tilde{\mathbf{p}}_{\text{rel}}(t_l)\| = \tilde{r}_{\text{tlz}}$ from the interior. If $T_{\text{lim}}^{\text{next}} = 0$ or $\|\tilde{\mathbf{v}}_{\text{rel}}(t_l)\| = 0$, then the switch from $h_2(\mathbf{x}, t)$ to $h_3(\mathbf{x})$ yields $h_3(\mathbf{x}(t_l)) \geq 0$. If $T_{\text{lim}}^{\text{next}} > 0$ and $\|\tilde{\mathbf{v}}_{\text{rel}}(t_l)\| > 0$, then the switch from $h_2(\mathbf{x}, t)$ to $h_1(\mathbf{x})$ yields $h_1(\mathbf{x}(t_l)) \geq 0$.

Proof. Since the boundary is reached from the TLZ interior, there exists $\epsilon > 0$ such that $\|\tilde{\mathbf{p}}_{\text{rel}}(t)\| < \tilde{r}_{\text{tlz}}$ for $t \in (t_l - \epsilon, t_l)$. We first show that the exit-boundary state satisfies $h_3(\mathbf{x}(t_l)) \geq 0$. If $h_3(\mathbf{x}(t_l)) < 0$, then

$$\left. \frac{d}{dt} \|\tilde{\mathbf{p}}_{\text{rel}}\|^2 \right|_{t=t_l} = 2\tilde{\mathbf{p}}_{\text{rel}}(t_l)^\top \tilde{\mathbf{v}}_{\text{rel}}(t_l) < 0 \quad (35)$$

Hence, for $t < t_l$ sufficiently close to t_l , we would have $\|\tilde{\mathbf{p}}_{\text{rel}}(t)\|^2 > \|\tilde{\mathbf{p}}_{\text{rel}}(t_l)\|^2 = \tilde{r}_{\text{tlz}}^2$. This contradicts the fact that the UAV is inside the TLZ immediately before t_l . Therefore, we have $h_3(\mathbf{x}(t_l)) \geq 0$.

If $T_{\text{lim}}^{\text{next}} = 0$ or $\|\tilde{\mathbf{v}}_{\text{rel}}(t_l)\| = 0$, the switch from $h_2(\mathbf{x}, t)$ to $h_3(\mathbf{x})$ therefore yields $h_3(\mathbf{x}(t_l)) \geq 0$ at the exit boundary. If $T_{\text{lim}}^{\text{next}} > 0$ and $\|\tilde{\mathbf{v}}_{\text{rel}}(t_l)\| > 0$, then at the exit boundary, we have $q = \frac{1}{4} \|\tilde{\mathbf{v}}_{\text{rel}}(t_l)\|^2 (T_{\text{lim}}^{\text{next}})^2$. Thus, $q > 0$ when $\|\tilde{\mathbf{v}}_{\text{rel}}(t_l)\| > 0$, and $h_1(\mathbf{x})$ is well defined at the reset boundary state. Moreover, $h_1(\mathbf{x}(t_l)) = h_3(\mathbf{x}(t_l)) + \frac{1}{2} \|\tilde{\mathbf{v}}_{\text{rel}}(t_l)\|^2 T_{\text{lim}}^{\text{next}} \geq 0$. Therefore, the switch from $h_2(\mathbf{x}, t)$ to $h_1(\mathbf{x})$ also yields $h_1(\mathbf{x}(t_l)) \geq 0$ at the exit boundary. $\square$

At an entry event handled by $h_2(\mathbf{x}, t)$, $\|\tilde{\mathbf{v}}_{\text{rel}}(t_s)\| > 0$ holds by the switching rule. Theorem 4 gives $h_2(\mathbf{x}(t_s), t_s) \geq 0$, and Theorem 2 guarantees its preservation while $h_2(\mathbf{x}, t)$ is active and its CBF condition is enforced. If $\tilde{\mathbf{v}}_{\text{rel}}(t) = \mathbf{0}$ at a strict interior state, then $\mathbf{s}(t) = \tilde{\mathbf{p}}_{\text{rel}}(t)$ and $h_2(\mathbf{x}(t), t) = \|\tilde{\mathbf{p}}_{\text{rel}}(t)\|^2 - \tilde{r}_{\text{tlz}}^2 < 0$, contradicting the maintained condition $h_2(\mathbf{x}(t), t) \geq 0$. Therefore, zero relative velocity cannot occur in the strict interior while $h_2(\mathbf{x}, t)$ is active. It remains to analyze the switches between $h_1(\mathbf{x})$ and $h_3(\mathbf{x})$, which arise at a strict-NFZ boundary or at zero relative velocity in $\mathcal{D}_3$.

Theorem 6 (Switching between h_1 and h_3). *Consider either of the following cases: (i) Strict-NFZ boundary: $T_{\text{lim}} = 0$, and the UAV reaches $\|\tilde{\mathbf{p}}_{\text{rel}}(t_{a_1})\| = \tilde{r}_{\text{tlz}}$ from the exterior with $h_1(\mathbf{x}(t)) \geq 0$ maintained immediately before t_{a_1} ; or (ii) Zero relative velocity: $\mathbf{x}(t_{a_2}) \in \mathcal{D}_3$ and $\|\tilde{\mathbf{v}}_{\text{rel}}(t_{a_2})\| = 0$. With $k \in \{1, 2\}$ denoting the case index, activating $h_3(\mathbf{x})$ at t_{a_k} yields $h_3(\mathbf{x}(t_{a_k})) \geq 0$. If the CBF condition for $h_3(\mathbf{x})$ is enforced until $t_{b_k} > t_{a_k}$, where $\mathbf{x}(t_{b_k}) \in \mathcal{D}_1$ and $\|\tilde{\mathbf{v}}_{\text{rel}}(t_{b_k})\| > 0$, then switching back to $h_1(\mathbf{x})$ yields $h_1(\mathbf{x}(t_{b_k})) \geq 0$.*

Proof. For case (i), we have $q(t_{a_1}) = 0$ when $\|\tilde{\mathbf{p}}_{\text{rel}}(t_{a_1})\| = \tilde{r}_{\text{tlz}}$. Although $h_1(\mathbf{x})$ is not activated there because it is not differentiable, its defining expression has the boundary value $\lim_{t \rightarrow t_{a_1}^-} h_1(\mathbf{x}(t)) = \tilde{\mathbf{p}}_{\text{rel}}(t_{a_1})^\top \tilde{\mathbf{v}}_{\text{rel}}(t_{a_1}) = h_3(\mathbf{x}(t_{a_1}))$. Since $h_1(\mathbf{x}(t)) \geq 0$ before the switch, this limit is nonnegative, and hence $h_3(\mathbf{x}(t_{a_1})) \geq 0$. For case (ii), $\tilde{\mathbf{v}}_{\text{rel}}(t_{a_2}) = \mathbf{0}$ directly gives $h_3(\mathbf{x}(t_{a_2})) = 0$.

By Theorem 3, enforcing the CBF condition for $h_3(\mathbf{x})$ preserves $h_3(\mathbf{x}) \geq 0$. At switching time t_{b_k} , $k \in \{1, 2\}$, the stated conditions ensure that $h_1(\mathbf{x})$ is differentiable and $q(t_{b_k}) > 0$. Therefore, $h_1(\mathbf{x}(t_{b_k})) = h_3(\mathbf{x}(t_{b_k})) + \|\tilde{\mathbf{v}}_{\text{rel}}(t_{b_k})\| \sqrt{q(t_{b_k})} \geq 0$. $\square$

Remark 3. *Although case (i) of Theorem 6 establishes switching safety at the exact TLZ boundary, small thresholds $\sigma > 0$ and $\nu > 0$ can be used in implementation for the boundary distance and relative speed, respectively. After $h_3(\mathbf{x})$ is selected, it remains active until both $\|\tilde{\mathbf{p}}_{\text{rel}}\| - \tilde{r}_{\text{tlz}} > \sigma$ and $\|\tilde{\mathbf{v}}_{\text{rel}}\| > \nu$ are satisfied; only then does the active component return to $h_1(\mathbf{x})$. This switching rule is valid because $h_3(\mathbf{x})$ is a valid CBF throughout $\mathcal{D}_3$ by Theorem 3. The two thresholds reduce repeated switching between $h_1(\mathbf{x})$ and $h_3(\mathbf{x})$.*

C. Dwell-Time Guarantee

Following the switching analysis, we show that maintaining $h_2(\mathbf{x}, t) \geq 0$ during a TLZ traversal guarantees that the dwell duration does not exceed T_{lim} .

Lemma 2 (Dwell-time guarantee under h_2). *Suppose that the UAV enters the TLZ at t_s with $T_{\text{rem}}(t_s) = T_{\text{lim}} > 0$ and $h_2(\mathbf{x}(t_s), t_s) \geq 0$. If the CBF condition associated with $h_2(\mathbf{x}, t)$ is enforced while the UAV remains inside the TLZ, then the dwell duration of this traversal does not exceed T_{lim} .*

Proof. Let $t^* = t_s + T_{\text{lim}}$. For $t \in [t_s, t^*)$, $T_{\text{rem}}(t) > 0$. By Theorem 2 and the standard CBF forward-invariance result, enforcing the CBF condition for $h_2(\mathbf{x}, t)$ gives $h_2(\mathbf{x}(t), t) \geq 0$ while the UAV remains inside the TLZ. If the UAV has not exited before t^* , forward invariance gives $h_2(\mathbf{x}(t), t) \geq 0$ for all $t < t^*$. Since $T_{\text{rem}}(t) \rightarrow 0$ as $t \rightarrow t^*-$ and $h_2(\mathbf{x}, t)$ is continuous in T_{rem} , it follows that $h_2(\mathbf{x}(t^*), t^*) \geq 0$. Since $T_{\text{rem}}(t^*) = 0$, we have $h_2(\mathbf{x}(t^*), t^*) = \|\tilde{\mathbf{p}}_{\text{rel}}(t^*)\|^2 - \tilde{r}_{\text{tlz}}^2$, which would be negative if the UAV were still strictly inside the TLZ. Therefore, the UAV must exit the TLZ before t^* or reach the TLZ boundary at t^* , so the dwell duration is no greater than T_{lim} . $\square$

Lemma 2 shows that maintaining $h_2(\mathbf{x}, t) \geq 0$ during a TLZ traversal is sufficient for the dwell-time guarantee of that

traversal. Based on this result, the following theorem establishes the overall dwell-time guarantee of the PLTCBF.

Theorem 7 (Satisfaction of the dwell-time constraint). *Consider the PLTCBF $h_{i_{\text{act}}}(\mathbf{x}, t)$ defined in Eq. (24). Suppose that the UAV starts outside the TLZ and that its initially active component is nonnegative. Then solving the CBF-QP in Eq. (12) guarantees satisfaction of the dwell-time constraint for every traversal.*

Proof. By assumption, the initially active component is non-negative. Theorems 1–3 establish that each component CBF is valid. Hence, the nonnegativity of the active component is preserved while it is active.

Theorems 4–6 ensure that the newly activated component is nonnegative at every switch. While $h_3(\mathbf{x})$ is active outside the TLZ, $h_3(\mathbf{x}) \geq 0$ gives $\frac{d}{dt} \|\tilde{\mathbf{p}}_{\text{rel}}\|^2 = 2h_3(\mathbf{x}) \geq 0$, so the UAV cannot enter the TLZ. If the relative velocity becomes nonzero, case (ii) of Theorem 6 permits a safe return to $h_1(\mathbf{x})$. Therefore, every traversal with $T_{\text{lim}} > 0$ begins through the entry switch of Theorem 4, which gives $h_2(\mathbf{x}(t_s), t_s) \geq 0$. Its nonnegativity is then preserved while the UAV remains inside the TLZ. Lemma 2 consequently guarantees that the dwell-time constraint is satisfied for each such traversal. If $T_{\text{lim}} = 0$, case (i) of Theorem 6 guarantees that $h_3(\mathbf{x}) \geq 0$ when the UAV reaches the TLZ boundary from the exterior. This inequality remains satisfied while $h_3(\mathbf{x})$ is active. Hence, $\frac{d}{dt} \|\tilde{\mathbf{p}}_{\text{rel}}\|^2 = 2h_3(\mathbf{x}) \geq 0$, and the UAV cannot enter the TLZ. Therefore, no dwell interval occurs when $T_{\text{lim}} = 0$. Combining the two cases completes the proof. $\square$

V. NUMERICAL SIMULATIONS

In this section, numerical simulations are conducted to evaluate the effectiveness of the proposed PLTCBF method under dwell-time constraints. In all simulations, the baseline trajectory is obtained by directly applying the nominal acceleration command $\mathbf{u}_{\text{nom}}$ without enforcing the CBF-QP safety filter. For multi-TLZ scenarios, an independent timer is maintained for each TLZ, and all currently active CBF constraints are imposed simultaneously in the CBF-QP.

Since TLZ entry may occur between samples, the implementation uses $T_{\text{lim}} - \Delta t$ while the UAV is outside the TLZ and initializes $T_{\text{rem}} = T_{\text{lim}} - \Delta t$ at the first inside sample. The entry and exit times reported below are obtained by linear interpolation between the samples immediately before and after the UAV crosses the TLZ boundary.

All MATLAB simulations in this section were conducted in MATLAB R2025b on a computer equipped with an Intel Core Ultra X7 358H processor and 32 GB RAM. The CBF-QP problems were solved using MATLAB `quadprog` with a sampling frequency of 100 Hz. Computation times were measured using MATLAB's `tic/toc` functions. These simulations use the acceleration bounds $\mathbf{u}_{\text{min}} = [-2, -2, -2]^\top$ m/s² and $\mathbf{u}_{\text{max}} = [2, 2, 2]^\top$ m/s². At each sampling instant, the nominal acceleration command is defined as

$$\mathbf{u}_{\text{nom}} = \text{sat}(K(\mathbf{v}_{\text{des}} - \mathbf{v})) \quad (36)$$

The desired velocity used in this command is $\mathbf{v}_{\text{des}} = v_c \tanh(0.05\|\delta\|)\delta/\|\delta\|$. Here, $\text{sat}(\cdot)$ limits the acceleration to

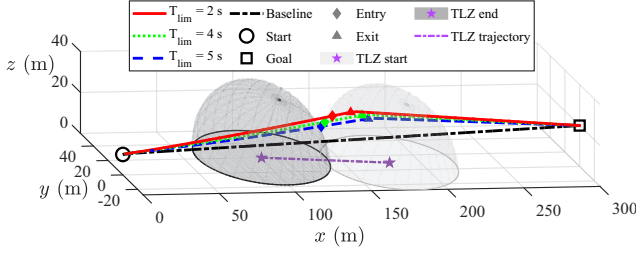


Fig. 3. Trajectories for different T_{lim} in the moving ellipsoidal TLZ scenario

the interval $[\mathbf{u}_{min}, \mathbf{u}_{max}]$, $K = 5$ for the MATLAB simulations, $\delta = \mathbf{p}_{ref} - \mathbf{p}$ is the vector from the UAV to the goal, and v_c denotes the desired cruising speed. Each simulation is terminated once the target-distance tolerance $\|\delta\| \leq 0.1$ m is reached.

The extended class $\mathcal{K}_\infty$ function is chosen as $\kappa(h) = kh$ with $k = 1$. For method comparison, the control correction is measured by $J_u = \int_0^{t_f} \|\mathbf{u}_{safety}(t) - \mathbf{u}_{nom}(t)\| dt$, where t_f is the termination time and the metric J_u has units of m/s.

A. Scenario with a Moving Ellipsoidal TLZ

The first scenario considers a single moving ellipsoidal TLZ. The UAV starts from $\mathbf{p}_0 = [0, 0, 10]^T$ m with $\mathbf{v}_0 = [0, 0, 0]^T$ m/s and moves toward $\mathbf{p}_{ref} = [300, 10, 20]^T$ m with desired cruising speed $v_c = 6$ m/s. The TLZ has initial center $[180, 10, 0]^T$ m, semi-axes $[42, 38, 35]^T$ m, rotation $[20^\circ, 10^\circ, 0^\circ]^T$, and velocity $\mathbf{v}_{tlz} = [-1.2, 0.15, 0]^T$ m/s. To evaluate the influence of the dwell-time limit, the simulations are conducted with $T_{lim} = 2$ s, 4 s, and 5 s.

Figs. 3 and 4 show that the proposed PLTCBF allows the UAV to temporarily traverse the moving ellipsoidal TLZ for different values of T_{lim} . In Fig. 4, the marked entry and exit times satisfy $t_{out} - t_{in} \leq T_{lim}$ for all tested values of T_{lim} . Fig. 5 shows that the acceleration commands satisfy the control saturation constraint in Eq. (6). Fig. 6 shows the corresponding three-axis velocity responses for the tested values of T_{lim} .

B. Scenario with a Moving Elliptic-Cylinder TLZ

The second scenario considers a moving elliptic-cylinder TLZ. The UAV starts from $\mathbf{p}_0 = [0, 0, 10]^T$ m with $\mathbf{v}_0 = [0, 0, 0]^T$ m/s and moves toward $\mathbf{p}_{ref} = [300, 10, 25]^T$ m with desired cruising speed $v_c = 6$ m/s. The TLZ has initial center $[180, 8, 0]^T$ m, horizontal semi-axes $[35, 30]^T$ m, rotation 30° , and velocity $\mathbf{v}_{tlz} = [-1, 0.1, 0]^T$ m/s. The simulations are conducted with $T_{lim} = 2$ s, 4 s, and 5 s.

Figs. 7 and 8 show that the proposed PLTCBF allows temporary traversal of the moving elliptic-cylinder TLZ for different values of T_{lim} . In Fig. 8, the marked entry and exit times satisfy $t_{out} - t_{in} \leq T_{lim}$ for all tested values of T_{lim} . Fig. 9 shows that the acceleration commands satisfy the control saturation constraint in Eq. (6). Since the elliptic-cylinder constraint is evaluated in the transformed horizontal plane, the main control variation appears in a_x and a_y . Fig. 10 presents the velocity profiles during the TLZ traversal for the tested values of T_{lim} . These results verify that the proposed PLTCBF

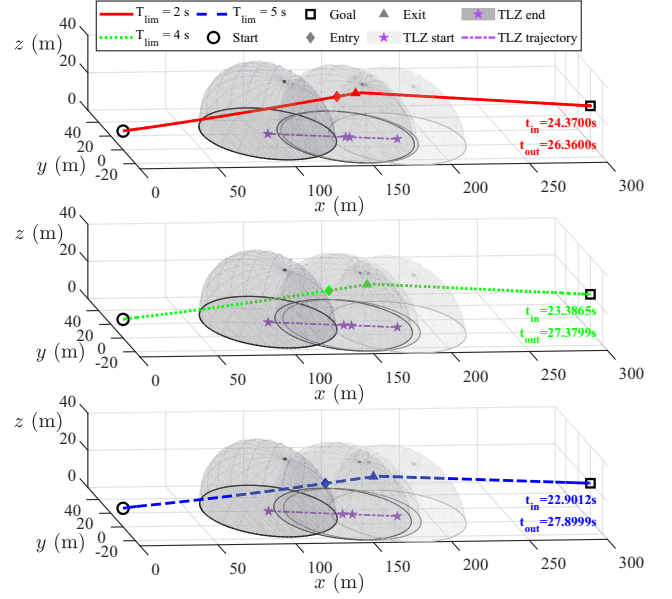


Fig. 4. Trajectories with TLZ entry and exit times for different T_{lim} in the moving ellipsoidal TLZ scenario

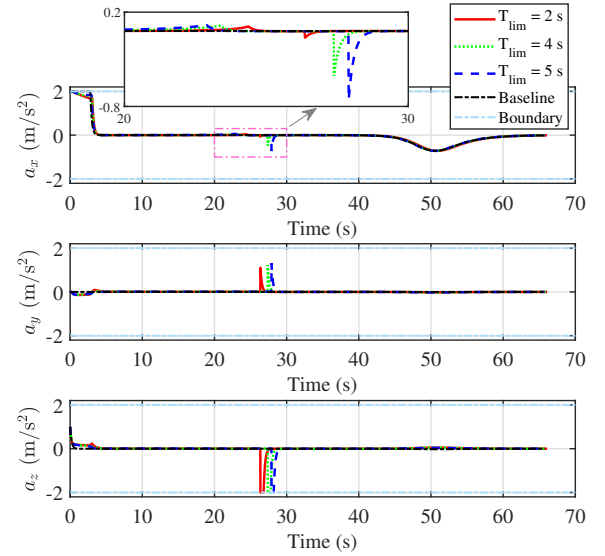


Fig. 5. Acceleration for different T_{lim} in the moving ellipsoidal TLZ scenario

can handle dwell-time constraints for elliptic-cylinder TLZs through the transformed horizontal plane.

C. Comparison Scenario with Multiple Ellipsoidal TLZs

The proposed PLTCBF method is compared with C3BF [31], HoCBF [32], nonlinear model predictive control (NMPC) [16], [15], and the unconstrained nominal controller in Eq. (36). C3BF and HoCBF treat the TLZs as strict NFZs, whereas NMPC enforces the dwell-time constraint through receding-horizon trajectory optimization. The NMPC controller uses a 5 s receding horizon with a 0.1 s update interval and is solved by MATLAB `fmincon` using SQP and the previous solution for warm starting. Its cost penalizes tracking error, control effort, and dwell-time violation. In this scenario, the UAV starts from

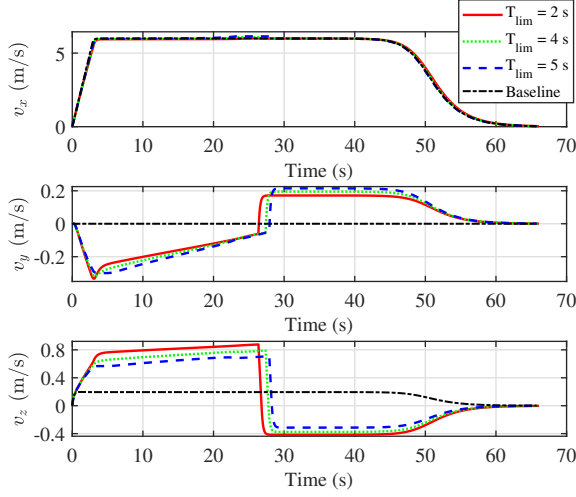


Fig. 6. Velocity for different T_{lim} in the moving ellipsoidal TLZ scenario

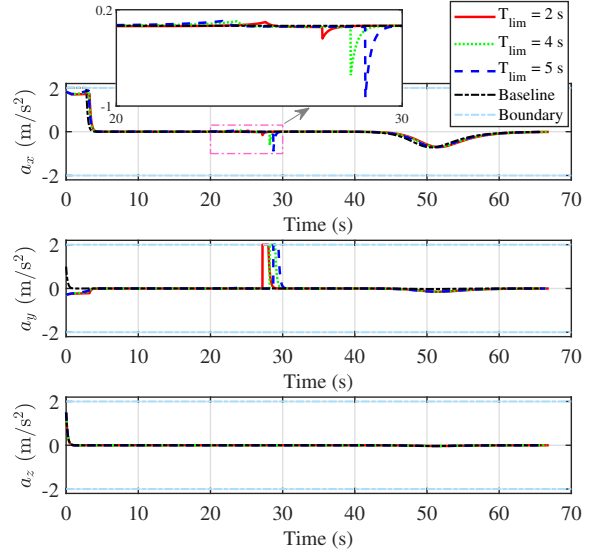


Fig. 9. Acceleration for different T_{lim} values in the moving elliptic-cylinder TLZ scenario

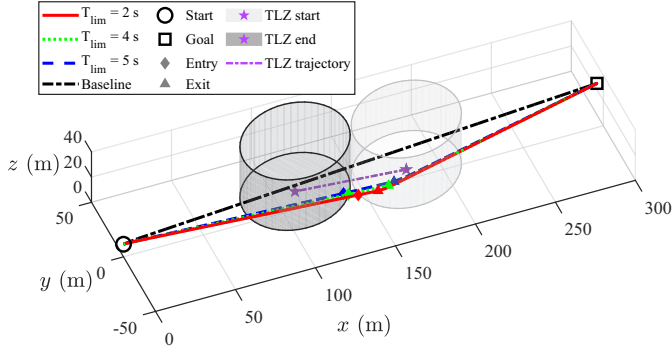


Fig. 7. Trajectories for different T_{lim} in the moving elliptic-cylinder TLZ scenario

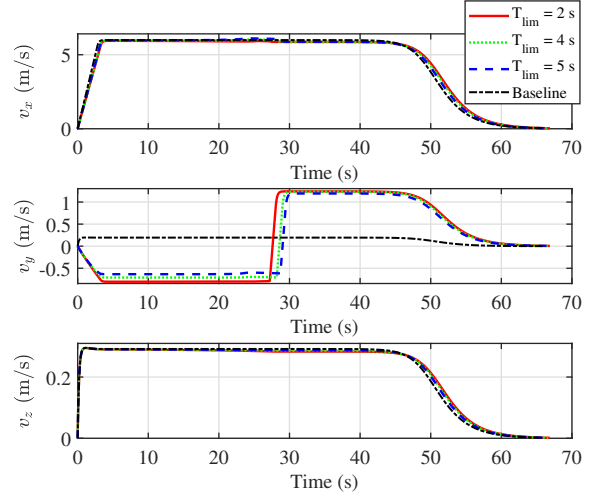


Fig. 10. Velocity for different T_{lim} values in the moving elliptic-cylinder TLZ scenario

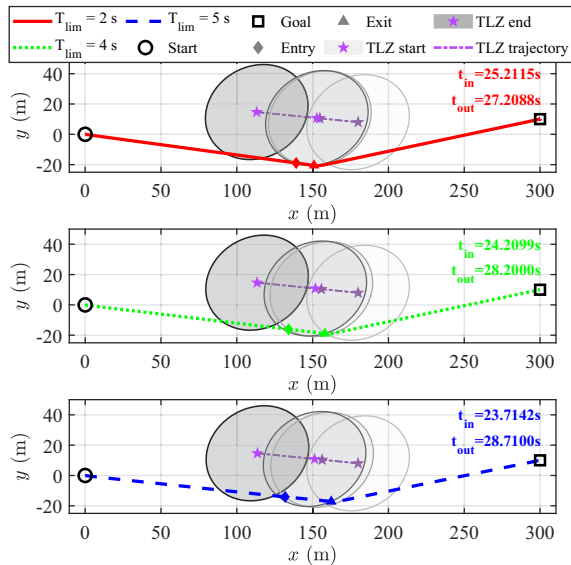


Fig. 8. Trajectories with TLZ entry and exit times for different T_{lim} in the moving elliptic-cylinder TLZ scenario

$p_0 = [0, 0, 5]^T$ m with $v_0 = [0.05, 0.01, 0]^T$ m/s and moves toward $p_{ref} = [200, 200, 10]^T$ m with desired cruising speed $v_c = 5$ m/s. Three ellipsoidal TLZs with $T_{lim} = 6$ s are considered, and their parameters are listed in Table I.

Fig. 11 shows that PLTCBF passes through TLZs 1 and 2 for 5.9925 s and 5.8604 s, respectively, satisfying the dwell-time constraints. NMPC likewise traverses TLZs 1 and 2, with dwell durations of 5.6543 s and 5.8294 s. In contrast, C3BF and HoCBF treat the TLZs as strict NFZs and therefore

TABLE I
PARAMETERS OF THE ELLIPSOIDAL TLZS

TLZ	Center (m)	Semi-axes (m)	Euler angles	Velocity (m/s)
1	$[100, 80, 0]^T$	$[32, 30, 30]^T$	$[0^\circ, 0^\circ, 0^\circ]^T$	$[0, 0, 0]^T$
2	$[60, 100, 0]^T$	$[32, 28, 30]^T$	$[0^\circ, 0^\circ, 0^\circ]^T$	$[0, 0, 0]^T$
3	$[60, 60, 35]^T$	$[25, 25, 15]^T$	$[0^\circ, 0^\circ, 0^\circ]^T$	$[0, 0, 0]^T$

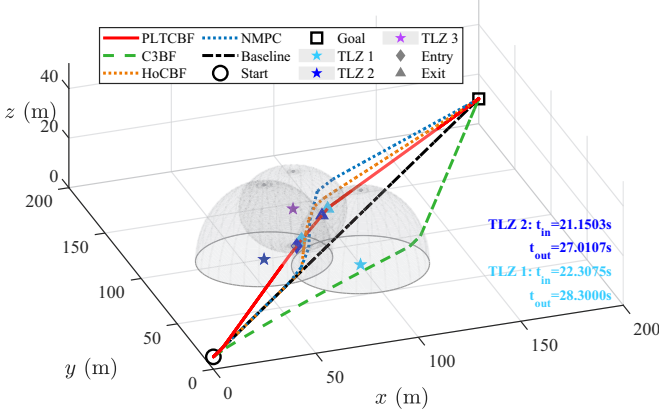


Fig. 11. Trajectory comparison of different methods in the scenario with multiple ellipsoidal TLZs

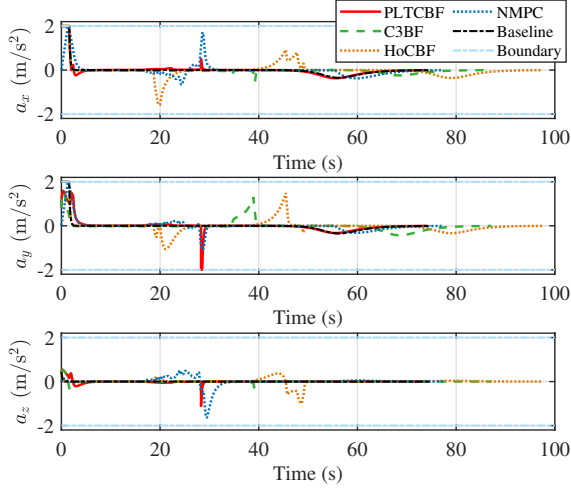


Fig. 12. Acceleration of different methods in the scenario with multiple ellipsoidal TLZs

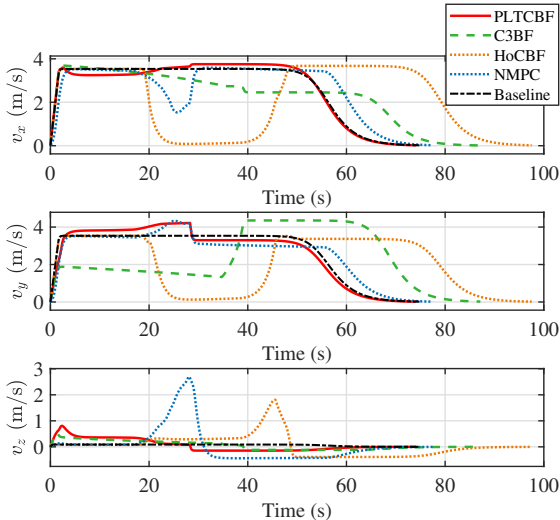


Fig. 13. Velocity of different methods in the scenario with multiple ellipsoidal TLZs

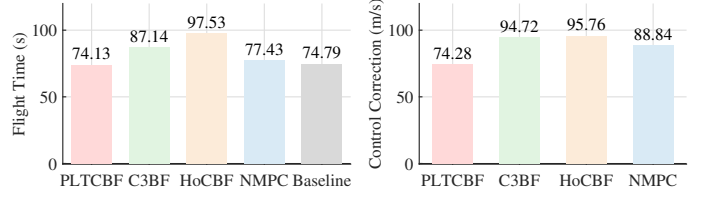


Fig. 14. Flight time and control correction of different methods in the scenario with multiple ellipsoidal TLZs

TABLE II
PARAMETERS OF THE ELLIPTIC-CYLINDER TLZs

TLZ	Center (m)	Semi-axes (m)	Rotation	Velocity (m/s)
1	$[80, 60, 0]^T$	$[20, 16]^T$	75°	$[0, 0, 0]^T$
2	$[60, 85, 0]^T$	$[25, 16]^T$	-18°	$[0, 0, 0]^T$
3	$[105, 50, 0]^T$	$[15, 12]^T$	0°	$[0, 0, 0]^T$

avoid them, resulting in longer detours. Although PLTCBF and NMPC both allow TLZ traversal while satisfying the dwell-time constraints, their per-solve computational costs differ markedly: NMPC requires 191.1 ms per optimization on average, whereas the PLTCBF CBF-QP requires only 0.7 ms per solve. Because C3BF and HoCBF likewise solve low-dimensional CBF-QPs, their mean per-solve times are similarly low at 0.7 ms and 0.8 ms, respectively. Overall, PLTCBF leverages the dwell-time constraint to permit temporary TLZ traversal while retaining a low computational burden, thereby improving mission efficiency.

The acceleration profiles in Fig. 12 show that all methods remain within the prescribed acceleration bounds, while PLTCBF requires comparatively modest control adjustments among the constrained methods. Figure 13 shows that the PLTCBF velocity profiles remain closer to the baseline. Figure 14 further shows that, among the constrained methods, PLTCBF achieves the shortest flight time and remains closest to the baseline. PLTCBF also yields the smallest cumulative control correction among the constrained methods, indicating that it requires the least modification of the nominal control while satisfying the dwell-time constraints.

D. Comparison Scenario with Multiple Elliptic-Cylinder TLZs

The same methods and controller configurations are next evaluated in a scenario with three stationary elliptic-cylinder TLZs. The UAV starts from $p_0 = [0, 0, 10]^T$ m with $v_0 = [0.02, 0.02, 0]^T$ m/s and is commanded toward $p_{ref} = [140, 120, 20]^T$ m at a desired cruising speed of $v_c = 5$ m/s. Each TLZ imposes a dwell-time limit of $T_{lim} = 4$ s. The centers, horizontal semi-axes, orientations, and velocities of the three TLZs are listed in Table II.

For the elliptic-cylinder configuration, Fig. 15 shows that PLTCBF and NMPC traverse TLZs 1 and 2 while satisfying the dwell-time requirement. The dwell durations are 3.9924 s and 3.8650 s for PLTCBF, and 3.8033 s and 3.9864 s for NMPC. In contrast, C3BF and HoCBF treat all TLZs as strict NFZs, and therefore avoid them by following longer detours. The mean solve time of PLTCBF is 0.7 ms, comparable to

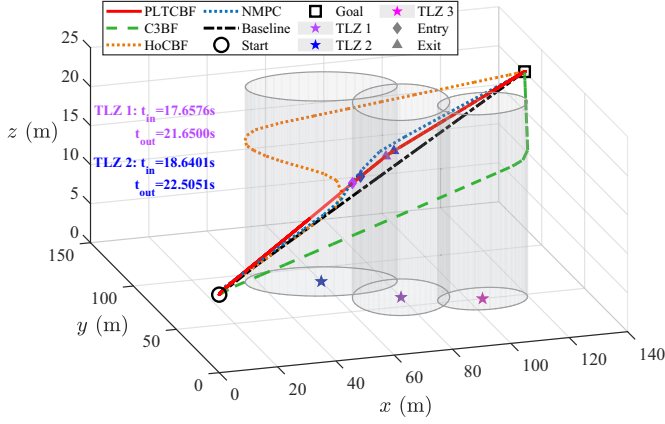


Fig. 15. Trajectories of different methods in the scenario with multiple elliptic-cylinder TLZs

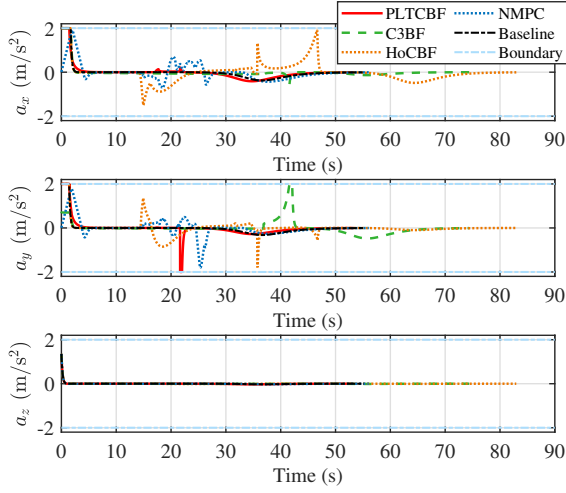


Fig. 16. Acceleration of different methods in the scenario with multiple elliptic-cylinder TLZs

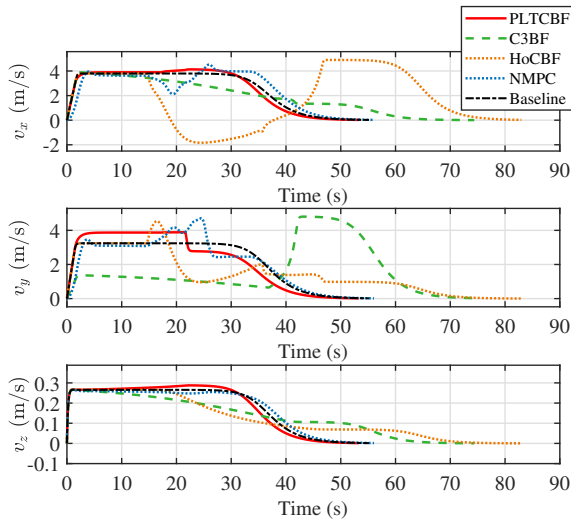


Fig. 17. Velocity of different methods in the scenario with multiple elliptic-cylinder TLZs

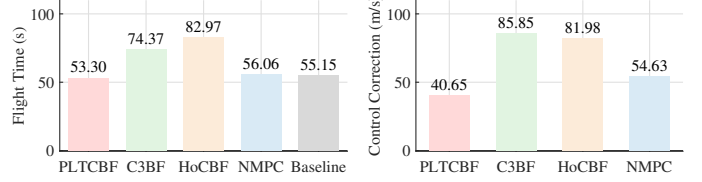


Fig. 18. Flight time and control correction of different methods in the scenario with multiple elliptic-cylinder TLZs

0.7 ms for C3BF and 0.8 ms for HoCBF, whereas NMPC requires 224.1 ms per optimization on average. Compared with NFZ-based CBF methods, PLTCBF handles the dwell-time constraint directly and reduces unnecessary detours; compared with NMPC, its low computational cost facilitates online implementation.

Figures 16 and 17 show that PLTCBF primarily adjusts the horizontal acceleration and velocity components to enable TLZ traversal, while the vertical responses remain close to the baseline. Compared with the other constrained methods, PLTCBF maintains smaller deviations from the baseline with fewer repeated fluctuations in the control and velocity responses. Figure 18 further shows that, among the constrained methods, PLTCBF achieves the shortest flight time and remains closest to the baseline. It also yields the smallest cumulative control correction.

VI. SIMULATION IN COPPELIASIM

A closed-loop quadrotor simulation is conducted in CoppeliaSim V4.9.0 to validate the implementation. The quadrotor starts from $\mathbf{p}_0 = [0, 0, 5]^\top$ m with $\mathbf{v}_0 = [0, 0, 0]^\top$ m/s and moves toward $\mathbf{p}_{\text{ref}} = [50, 50, 15]^\top$ m with desired cruising speed $v_c = 2$ m/s. The moving elliptic-cylinder TLZ has initial horizontal center $[35, 30]^\top$ m, horizontal semi-axes $[10, 8]^\top$ m, orientation 20° , and constant horizontal velocity $\mathbf{v}_{\text{tlz}} = [-0.4, -0.2]^\top$ m/s. The dwell-time limit is $T_{\text{lim}} = 5$ s, and the acceleration bounds are $\mathbf{u}_{\text{min}} = [-0.5, -0.5, -0.5]^\top$ m/s² and $\mathbf{u}_{\text{max}} = [0.5, 0.5, 0.5]^\top$ m/s².

At each sampling instant, the simulated position $\mathbf{p}_k$ and measured velocity $\mathbf{v}_{\text{actual},k}$ are used to calculate $\mathbf{u}_{\text{nom},k}$ according to Eq. (36) with $K = 1$, and the PLTCBF filter returns $\mathbf{u}_{\text{safety},k}$. This command is converted into a resultant force and then into propeller commands by the thrust and attitude controllers. The actual acceleration is estimated as $\mathbf{u}_{\text{actual},k} = (\mathbf{v}_{\text{actual},k} - \mathbf{v}_{\text{actual},k-1})/\Delta t$, with $\Delta t = 0.05$ s, and the simulation stops once the target tolerance is reached. In the velocity plot, $\mathbf{v}_{\text{des},k}$ is calculated using the desired-velocity law above, and $\mathbf{v}_{\text{safety},k} = \mathbf{v}_{\text{actual},k} + \mathbf{u}_{\text{safety},k}\Delta t$ is the one-step velocity prediction under the safety-filtered command.

Figure 19 shows that the quadrotor traverses the elliptic-cylinder TLZ and reaches the target. The entry and exit times are $t_{\text{in}} = 18.5778$ s and $t_{\text{out}} = 23.5499$ s, respectively, yielding a dwell duration of 4.9721 s and satisfying $T_{\text{lim}} = 5$ s. In Fig. 20, the actual acceleration $\mathbf{u}_{\text{actual}}$ closely follows the safety-filtered command $\mathbf{u}_{\text{safety}}$ and remains within the acceleration bounds. Figure 21 shows close agreement between $\mathbf{v}_{\text{actual}}$ and $\mathbf{v}_{\text{safety}}$, with $\mathbf{v}_{\text{des}}$ providing the nominal reference.

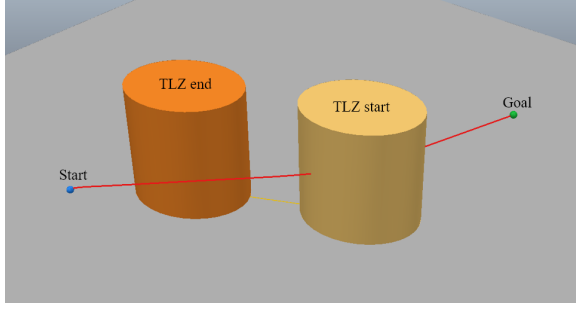


Fig. 19. UAV trajectory in CoppeliaSim

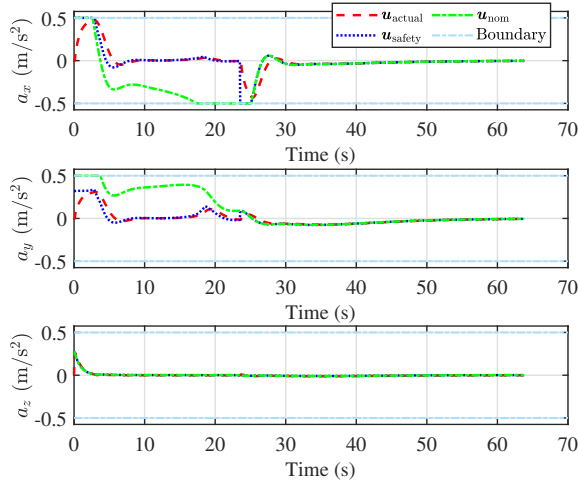


Fig. 20. Acceleration in the CoppeliaSim simulation: u_{nom} , u_{safety} , and u_{actual}

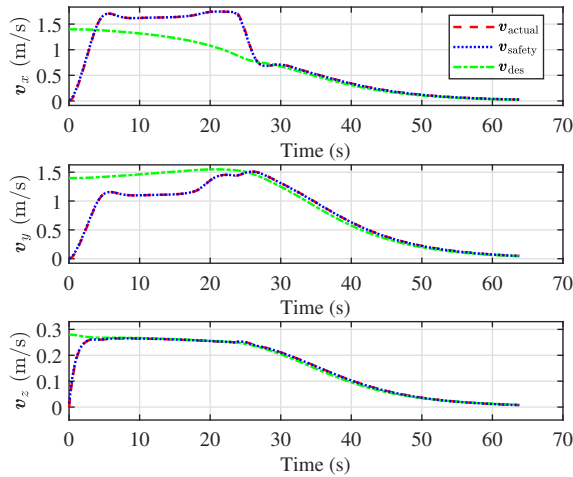


Fig. 21. Velocity in the CoppeliaSim simulation: v_{actual} , v_{des} , and v_{safety}

These results confirm that PLTCBF can be incorporated into the closed-loop quadrotor control architecture while enforcing the dwell-time constraint.

VII. CONCLUSION

This paper developed a PLTCBF method for enforcing dwell-time constraints induced by TLZs in 3D UAV flight. The PLT was constructed from the current state and used to characterize the dwell-time constraint through the geometric relationship between the PLT and the TLZ in the transformed space. For ellipsoidal and elliptic-cylinder TLZs, coordinate transformations were used to obtain tractable TLZ representations, which led to explicit CBF constructions. The active component CBF was selected according to the UAV-TLZ relation, and the analysis showed that its nonnegativity is preserved across TLZ-boundary and zero-relative-velocity switches, leading to the prescribed dwell-time guarantee under the CBF-QP implementation.

Numerical simulations under different dwell-time limits and TLZ configurations demonstrate the effectiveness of PLTCBF in satisfying the dwell-time constraints. Compared with C3BF and HoCBF, which conservatively treat TLZs as strict NFZs, PLTCBF reduces unnecessary detours and achieves shorter flight times with smaller cumulative control corrections. Its CBF-QP retains millisecond-level solution times and requires substantially less computation than NMPC, while both methods are able to exploit the allowable dwell time to traverse TLZs. Finally, the closed-loop CoppeliaSim simulation shows that the safety-filtered commands can be followed within the quadrotor control loop while the dwell-time constraint remains satisfied.

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